Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/15/2006
PI Name:	Anderson, Tim Ph.D.		
Project Title:	An Electrochemical Method to Visualize Flow in Liquid Metals		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Gainesville	**State:**	FL
Zip Code:	32611	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/29/2007
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Banks, Curtis	**Contact Phone:**	256 544-4437
Contact Email:	curtis.e.banks@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/29/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Narayanan, Ranga (University of Florida)		
Grant/Contract No.:	NNM04AA27G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of this ground-based research program is to extend and apply solid state electrochemical titration procedures to identify the dynamic convective states in liquid metals that can occur, for example, when density gradients are present. Density gradients can exist during liquid metal processing as a result of solutal or thermal gradients and the subsequent flow can substantially influence processed material properties. Previous research has developed and validated a visualiztion system appropriate for contained liquid metal systems. This system will be applied visualization of flow during growth of a pseudobianry compound semiconductor system. In addition an integrated microsensor array prototype fabricated using lithographic techniques will be developed. A set of applications will be explored.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The major contributions of this study have been: 1. The oxygen diffusivity of liquid tin and liquid tin-lead alloys has been measured as a function of temperature over a large temperature range. 2. The second critical Rayleigh numbers were computed as a function of Cartesian aspect ratio for one of the lowest Prandtl number fluids ever investigated. 3. The first critical Rayleigh number was detected using CFD modeling and concentration gradient applied over diffusivity ampoules. 4. The use of the flow visualization sensor using YSZ solid electrolytes was demonstrated by detecting oscillatory flow in a cylindrical alumina enclosure.
Bibliography Type:	Description: (Last Updated:)